Main Board Architecture & Manufacturing Report

Topic: Hardware Redesign (Post-MAX77642) & PCBA Vendor Quotation Analysis

Date: April 2026

1. Executive Summary

Due to global supply chain shortages and procurement difficulties regarding the **MAX77642ANA+** Power Management IC (PMIC), a hardware redesign of the main board's power distribution network was executed. The redesign successfully replaces the integrated PMIC with a robust, discrete power architecture ensuring better availability, higher current headroom, and cost-effective assembly.

Additionally, this report outlines the final Bill of Materials (BOM) adjustments for automated PCBA and compares quotes across three major manufacturers: JLCPCB, PCBWay, and NextPCB.

2. Hardware Architecture Modifications

The removal of the MAX77642ANA+ required a complete routing and schematic overhaul for the 3.3V and 1.8V power rails.

2.1. 3.3V Rail: TPS63021DSJR (Buck-Boost Converter)

- **Previous Limit:** ~1.0A (Configured via external resistors).
- **New Implementation:** The **TPS63021DSJR** replaces the MAX PMIC for the 3.3V rail.
- **Engineering Advantages:** It features an internal fixed hardware limit, capable of delivering **up to 2.0A continuously** (with 4A peak switch current). This provides massive headroom considering the 3.3V network's nominal draw is ~156mA. It includes automatic short-circuit protection.
- **Routing Updates:** Power Save / Sync pin (PS/SYNC) was routed directly to GND to enforce high-efficiency power-saving mode under light loads, maximizing the 26650 battery life.

2.2. 1.8V Rail: XC6206 (LDO Regulator)

- **New Implementation:** The **XC6206** (SOT-23-3) was cascaded from the new 3.3V output to generate the stable 1.8V required by the system logic.
- **Engineering Advantages:** Capable of 200mA output with internal foldback current limiting. Given the micro-amp level consumption of the 1.8V rail (~1.2mA), this LDO operates

seamlessly with negligible thermal dissipation.

2.3. Passive Component Adjustments

- **Removed:** The 0.47uH inductor (IHLP2020BZERR47M01) associated with the MAX IC was deleted.
- **Added/Modified:** The 1.5uH inductor count (NRS6028T1R5NMGJ) was increased. New dedicated filter network capacitors (10uF, 22uF, 1uF, and 0.1uF) were reassigned and routed, separating Analog Ground (AGND) for logic and Power Ground (PGND) for high-current loops to prevent electrical noise.

3. Vendor Quotation Analysis

The updated design BOMs were submitted to three major PCBA houses. The table below outlines the comparative pricing based on the latest redesign. *Note: Prices are based on a 5-unit prototype batch.*

Factory	**PCB Cost**	**Assembly**	**Components**	**Total (Not including shipping)**	**Estimated time**
JLCPCB	$135.80	$437.61	$50.64	**$624.05**	15 days
NextPCB	$141.06	$541.48	$304.49	**$987.03**	35 days
PCBWay	$393.05	$380.36	$307.13	**$1080.54**	30 days

Vendor Details & Assembly Notes:

- **JLCPCB (Primary Choice):** Features the lowest total cost and the fastest turnaround time. However, there are **5 parts missing** that require a mandatory pre-order into their Global Sourcing parts library before the final assembly order can be triggered.
- **NextPCB:** No major component shortages flagged, but it presents a significantly higher assembly cost and the longest estimated lead time (35 days).
- **PCBWay:** Stock is available and it offers competitive individual component pricing. However, the raw PCB manufacturing cost is notably higher than competitors, leading to the highest total project cost.

4. Strategic Recommendations & Next Steps

1. **Component Pre-ordering (Action Required):** If proceeding with JLCPCB, the 5 unavailable components must be purchased immediately via their Global Sourcing portal

to avoid assembly delays.

2. **Future DNP Mitigation:** While setting the battery holder to DNP solves current prototyping hurdles, for scalable mass production (V2.0+), we recommend replacing the plastic battery holder with **SMD metallic battery clips** (e.g., Keystone equivalents). This will allow 100% automated Pick & Place assembly, eliminating all manual labor bottlenecks.
3. **Documentation:** A separate "DNP file" will be generated to track all unpopulated components for the manual assembly phase.